Project Definition: Brain Training BookStore

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Project: Brain Training Bookstore
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Problem
We need a program to help manage a business that sells products. The business is a bookshop that works completely online. The program will help manage the stock, availability, and prices of different books. It will also record payments and track the delivery process, so customers feel safe. The goal is to make it easier to buy books in the city of Quito through a program.

Overview:
In the domain of online book sales and local delivery, customers often need a convenient way to find available books, compare prices, and place orders. For a small bookstore in Quito that manages sales and deliveries directly, customers currently need to communicate with the bookstore to ask about available books and arrange their purchases and deliveries. When a customer places an order, the bookstore is also responsible for delivering the books to the customer's location. As the number of customers and orders increases, having a simple way to manage the purchasing and delivery process becomes increasingly important. This project focuses on providing customers with a digital platform where they can browse the bookstore's catalog, view book prices, select and purchase books, and track the status and location of their delivery.

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Background
In recent years, book sales have been increasing and evolving. Thanks to the internet—along with changing reading habits—new ways of obtaining our favorite books have emerged, one of which is digital sales. Many people prefer to buy books online because it's easy to find a wide variety of titles from anywhere in the world, on any device, and because they can have the products delivered directly to their homes.

Despite the growth of online shopping, local brick-and-mortar bookstores face several limitations, which restrict readers' access to certain books, new releases, or international editions. On the other hand, large global platforms for selling e-books often lack a literary identity, reader-focused customer service, and delivery times optimized at the local and regional levels.

The creation of this digital bookstore, which offers online sales and home delivery, aims to connect readers with a diverse selection of books through a fast, secure, and personalized shopping experience.

System Lógica and Data Specifications

To accurately track inventory, process customer purchases, calculate order totals, and monitor delivery logistics within Quito, the system requires structured data records and predefined formulas.

Purchase & Delivery History Record Specification

Each transaction record in the system contains the following data attributes:

- **Book ISBN:** A short string identifying the book (e.g., "978-0132350884").
- **Quantity (Q):** An integer representing the number of book units ordered.
- **Unit Price (UP):** An integer representing the price of a single book unit in cents (e.g., 1000 represents $10.00, maximum allowed $25.00 / 2500 cents).
- **Order Date/Time (OT):** An integer representing the time of order placement in minutes since January 1, 2026.
- **Base Delivery Fee (DF):** An integer representing the base delivery charge in cents within Quito (e.g., 250 represents $2.50).
- **Distance Factor (K):** A decimal value representing the delivery distance multiplier based on the Quito destination zone (e.g., 1.0 for Central, 1.20 for North/South, 1.50 for Valleys).
- **Delivery Status (DS):** A string code representing the status ("PENDING", "IN_TRANSIT", "DELIVERED").
- **Delivery Date/Time (DT):** An integer representing the time of completed delivery in minutes since January 1, 2026.

Data Calculation Formulas

The total delivery fee (TDF) for an order is calculated as:

```
TDF = DF * K
```

The total purchase amount (TPA) for an order including delivery and taxes is computed as:

```
TPA = (Q * UP) + TDF
```

The fulfillment duration in days (FD) for a completed order is computed as:

```
FD = (DT - OT) / (24 * 60)
```

Sample Data & Metrics Tables

Table 1 - Sample Book Order & Delivery History

ISBN	Quantity (Q)	Order Date/Time (OT)	Unit Price (UP)	Base Fee (DF)	Distance Factor (K)	Delivery Date/Time (DT)
978-0132350884	2	5256000	$10.00	$2.50	1.00	5257440
978-0201616224	1	5258880	$15.00	$2.50	1.20	5260320
978-0134685991	3	5261760	$8.50	$2.50	1.50	5264640
978-0321356680	1	5263200	$22.00	$2.50	1.00	5264640

Table 2 - Computed Values for Sample Orders

ISBN	Order Date/Time (OT)	Total Delivery Fee (TDF)	Total Purchase Amount (TPA)	Fulfillment Duration (FD)
978-0132350884	5256000	$2.50	$22.50	1.00 day
978-0201616224	5258880	$3.00	$18.00	1.00 day
978-0134685991	5261760	$3.75	$29.25	2.00 days
978-0321356680	5263200	$2.50	$24.50	1.00 day

Performance & Operations Summary Metrics

To evaluate store operations and delivery effectiveness in Quito, the system tracks the following key performance indicators:

- **Total Revenue (TR):** Sum of all Total Purchase Amounts (TPA) across processed orders.
- **Average Fulfillment Time (AFT):** The average Fulfillment Duration (FD) in days for all delivered orders.
- **Category Fulfillment Rate (CFR):** The number of orders successfully delivered per book category per day.

Table 3 - Example Bookstore Operational Metrics

Total Revenue (TR)	**Total Orders**	**Average Fulfillment Time (AFT)**
$94.25	4	1.25 days